Mathematics Bootcamp

General Equilibrium and the Social Planner

Miguel Vázquez-Carrero

Compilation Date: September 1, 2026

Today

- **The household:** an infinite path, boundary conditions, shadow values
- **The Euler equation**, and the firm problem
- **Competitive equilibrium:** definition, characterization, and market clearing
- **Planner:** efficiency, supporting prices, and existence

From a Point to a Path

Day 4 studied a scalar choice

$$x \in D \subseteq \mathbb{R}.$$

Today the choice is an infinite sequence:

$$z = (x_0, x_1, x_2, \ldots).$$

In the model that follows, a candidate path satisfies one Euler equation at every date and one transversality condition at infinity.

Dated Commodities

A commodity is identified not only by its physical features, but also by its date and state. In this deterministic model, date is the key label.

Different Date, Different Good

Consumption at t and $t + 1$ are distinct commodities. Saving exchanges the current commodity for a future one; the interest rate will determine their relative price.

Two Market Structures

Arrow–Debreu

- Markets open only at date 0.
- Every dated good is traded there, at prices $\{p_t\}$.
- A single lifetime budget constraint.
- Perfect commitment: contracts signed at 0 are honored.

Sequential Markets

- Markets reopen at every date.
- Spot goods and one-period assets, at prices $\{w_t, r_t\}$.
- One budget constraint per date.
- Debt must be bounded by a no-Ponzi condition.

Both assume perfect information and perfect competition, and both deliver the same consumption allocations. We work with sequential markets.

The Environment

Dynamic Production Economy

The environment is

$$E = \left(\mathbb{N}_0, \beta, u, f, (A_t)_{t=0}^{\infty}, k_0\right).$$

These primitives are fixed before prices and allocations are determined. They define the **neoclassical growth model**, our canonical example for the rest of the day.

Primitives

- Dates $t = 0, 1, \ldots$
- Preferences: β and u
- Technology: f and A_t

Initial Resources

- The household owns $a_0 = k_0 > 0$.
- Labor supply is one unit each date.
- Capital depreciates at rate $\delta \in (0, 1]$.

Lifetime Utility

Lifetime Utility

For $c = (c_0, c_1, c_2, \ldots)$,

$$U(c) = \sum_{t=0}^{\infty} \beta^t u(c_t), \qquad 0 < \beta < 1.$$

The discount factor β gives the relative weight on later utility. We assume that this sum is well-defined and finite on admissible paths.

Monotonicity

$$u'(c_t) > 0$$

More consumption is preferred.

Strict concavity

$$u''(c_t) < 0$$

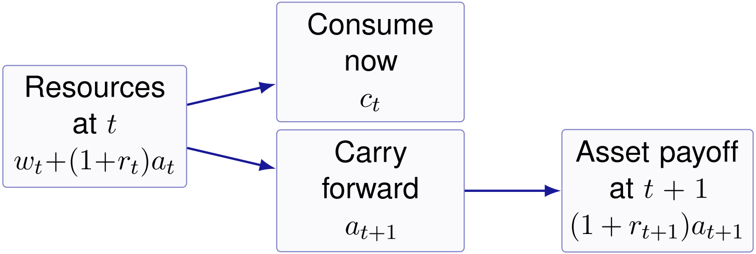
Marginal utility falls with consumption.

Saving Transfers Resources Across Dates

At date t, the household owns assets a_t, supplies one unit of labor, and receives wage and capital income.

Period Budget Constraint

$$c_t + a_{t+1} = w_t + (1 + r_t)a_t.$$



The Household Problem

Given $\{w_t, r_t\}_{t=0}^{\infty}$ and a_0

$$\max_{\{c_t, a_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to, for every t,

$$c_t + a_{t+1} = w_t + (1 + r_t)a_t, \qquad c_t \geq 0.$$

State and Control

The **state** a_t is predetermined at date t; the **control** c_t is chosen there. The budget constraint is the law of motion $a_{t+1} = w_t + (1 + r_t)a_t - c_t$.

Boundary Conditions

An Infinite Path Needs Both Ends

$$a_0 \text{ given,} \qquad \lim_{t\to\infty} \frac{a_{t+1}}{\prod_{i=1}^{t}(1+r_i)} = 0.$$

Where the Terminal Condition Comes From

No Ponzi, imposed on the problem: $\lim a_{t+1}/\prod(1+r_i) \geq 0$. Without it the household rolls debt over forever and no maximum exists.

Transversality, implied by optimality: $\lim a_{t+1}/\prod(1+r_i) \leq 0$. Leaving value unspent is never optimal.

Shadow Values

There is one budget constraint at each date, so the Lagrangean carries one multiplier at each date:

$$L = \sum_{t=0}^{\infty} \beta^t u(c_t) + \sum_{t=0}^{\infty} \lambda_t \Big[\underbrace{w_t + (1+r_t)a_t}_{\text{resources at } t} - c_t - a_{t+1} \Big].$$

What λ_t Measures

λ_t is the shadow value of wealth at date t: the gain in lifetime utility from one more unit of the good available at t. Differentiating in c_t gives

$$\beta^t u'(c_t) = \lambda_t.$$

The State Across Two Dates

The state a_{t+1} is chosen at t and carried into $t+1$, so it appears in two constraints:

$$\underbrace{c_t + a_{t+1} = w_t + (1+r_t)a_t,}_{\text{date } t,\text{ multiplier } \lambda_t} \qquad \underbrace{c_{t+1} + a_{t+2} = w_{t+1} + (1+r_{t+1})a_{t+1}}_{\text{date } t+1,\text{ multiplier } \lambda_{t+1}}.$$

Setting $\partial L/\partial a_{t+1} = 0$ therefore couples the two shadow values:

$$\lambda_t = (1+r_{t+1})\,\lambda_{t+1}.$$

Why This Is the Euler Equation

Saving one unit costs λ_t today and returns $(1+r_{t+1})\lambda_{t+1}$ tomorrow. Substituting $\lambda_t = \beta^t u'(c_t)$ eliminates the multipliers.

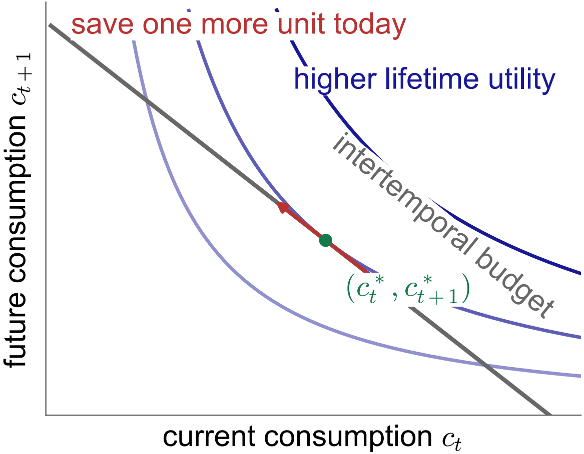
Tangency Across Dates

The household treats consumption at t and $t+1$ as two different goods. At an interior optimum,

$$\frac{\beta u'(c_{t+1})}{u'(c_t)} = \frac{1}{1+r_{t+1}}.$$

Same Mathematics as Day 4

The MRS equals the relative price. The only new feature is that the goods carry date labels.



Saving One More Unit: Marginal Cost Equals Marginal Benefit

Combine the two first-order conditions:

$$\boxed{u'(c_t) = \beta(1 + r_{t+1})u'(c_{t+1}).}$$

Cost of Saving One More Unit

One unit is forgone today, reducing lifetime utility by

$$\beta^t u'(c_t).$$

Benefit One Date Later

Tomorrow it yields $1 + r_{t+1}$ units. Lifetime utility rises by

$$\beta^{t+1}(1 + r_{t+1})u'(c_{t+1}).$$

At an interior optimum, these two marginal effects are equal.

The Euler Equation and the Tail

For the interior candidate path in this model, impose the model-specific transversality condition

$$\boxed{\lim_{t\to\infty} \beta^t u'(c_t) a_{t+1} = 0.}$$

Same condition as on *Boundary Conditions*: $\lambda_t = \beta^t u'(c_t)$ and $\lambda_t = \lambda_0 / \prod_{i=1}^{t}(1+r_i)$, so the two limits differ only by the constant λ_0.

Euler Equation

Compares the marginal cost and benefit of moving one unit between two adjacent dates.

Transversality Condition

Rules out tails that retain resources with positive shadow value.

Neither condition proves existence. Feasibility, concavity, and a separate existence argument remain necessary.

Technology

In every date t, the firm rents capital k_t, hires labor n_t, and produces the single final good:

$$y_t = A_t f(k_t, n_t) = A_t k_t^{\alpha} n_t^{1-\alpha}, \qquad 0 < \alpha < 1.$$

Technology

A_t is total factor productivity. It scales output at every input pair; f remains the standard Cobb–Douglas function.

Constant Returns

Under constant returns, any finite profit-maximizing plan earns zero profit. Positive profit would scale without bound.

The Firm Problem

Given w_t and r_t, at each date separately

$$\max_{k_t, n_t \geq 0} \left\{ k_t^{\alpha} (A_t n_t)^{1-\alpha} - w_t n_t - (r_t + \delta) k_t \right\}.$$

Static, and Prices Equal Marginal Products

There is no intertemporal link: the firm rents, produces and sells within date t. At an interior optimum,

$$w_t = A_t f_2(k_t, n_t), \qquad r_t + \delta = A_t f_1(k_t, n_t).$$

Factor Prices and the Return to Capital

For Cobb–Douglas production,

$$w_t = (1-\alpha)A_t k_t^{\alpha} n_t^{-\alpha}, \qquad r_t + \delta = \alpha A_t k_t^{\alpha-1} n_t^{1-\alpha}.$$

A fraction δ of the machine wears out in production, so the rental must cover depreciation on top of the net return. Save one unit today, receive $1 + r_t$ tomorrow.
Constant returns and the first-order conditions imply

$$A_t f(k_t, n_t) = w_t n_t + (r_t + \delta) k_t,$$

so competitive profits are zero.

Feasible Allocations

Allocation and Feasibility

A physical allocation is a sequence

$$z = (c_t, k_{t+1}, n_t)_{t=0}^{\infty}.$$

It is **feasible** if, for every $t = 0, 1, \ldots$,

$$c_t \geq 0, \qquad k_{t+1} \geq 0, \qquad 0 \leq n_t \leq 1,$$

and

$$\boxed{c_t + k_{t+1} \leq A_t k_t^{\alpha} n_t^{1-\alpha} + (1-\delta) k_t.}$$

Feasibility is physical and contains no prices. Monotonicity makes the resource inequality bind at an optimum.

Correspondences

Household Correspondence

At prices (w, r), write

$$H(w, r) = \{\text{household-optimal paths}\}.$$

It may contain more than one path (c, a).

Firm Correspondence

At prices (w_t, r_t), write

$$F_t(w_t, r_t) = \{\text{optimal input pairs at } t\}.$$

It may contain more than one pair (k, n).

Under constant returns,

$$\text{positive profit} \Longrightarrow F_t(w_t, r_t) = \varnothing, \qquad \text{zero profit} \Longrightarrow F_t(w_t, r_t) \text{ may be set-valued.}$$

Equilibrium selects mutually compatible elements of H and F_t.

Equilibrium: The General Definition

The Shape of Any Equilibrium Definition

Given the primitives and the initial state, an equilibrium is a list of *quantities* and *prices* such that

1. households **optimize**, taking prices as given;
2. firms **optimize**, taking prices as given;
3. markets **clear**.

A definition *names* the objects and *states* the conditions. It uses no derivatives, and it does not claim that such a list exists.

It is a *test*, not an adjustment story: it says which lists qualify, not how an economy would find one.

Equilibrium: The Actual Definition

The general definition, filled in for the growth model.

Competitive Equilibrium

Given a_0, a competitive equilibrium consists of household allocations $\{c_t, a_{t+1}\}_{t=0}^{\infty}$, firm allocations $\{k_t, n_t\}_{t=0}^{\infty}$ and prices $\{w_t, r_t\}_{t=0}^{\infty}$ such that

1. given $\{w_t, r_t\}$, the household allocation solves the **household problem**;
2. given $\{w_t, r_t\}$, the firm allocation solves the **firm problem** at every date;
3. **markets clear** at every date.

The firm pays $r_t + \delta$ per unit of capital, so the wage and the interest rate describe the whole price system.

Market Clearing

Three Markets, Every Date

$$\underbrace{a_t = k_t}_{\text{capital}}, \qquad \underbrace{n_t = 1}_{\text{labor}}, \qquad \underbrace{c_t + k_{t+1} = k_t^{\alpha}(A_t n_t)^{1-\alpha} + (1-\delta)k_t}_{\text{goods}}.$$

Walras' Law

If both problems are solved and two of the three markets clear, the third clears automatically. Goods-market clearing is implied by the other two.

Equilibrium: Characterization

Definition versus Characterization

The definition *states* conditions: solve, solve, clear.

The characterization *derives* what the conditions imply: the first-order conditions of each problem, joined to the market-clearing equations.

This is where derivatives enter, and nowhere earlier.

Existence and uniqueness are separate questions; a characterization settles neither.

The Equilibrium System

Every Equilibrium Path Satisfies, at Every Date

household	$u'(c_t) = \beta(1 + r_{t+1})u'(c_{t+1}),$	$c_t + a_{t+1} = w_t + (1 + r_t)a_t,$
firm	$w_t = (1 - \alpha)A_t k_t^{\alpha} n_t^{-\alpha},$	$r_t + \delta = \alpha A_t k_t^{\alpha-1} n_t^{1-\alpha},$
markets	$a_t = k_t,$	$n_t = 1.$

Six equations each date for the six unknown sequences $(c_t, a_{t+1}, k_t, n_t, w_t, r_t)$. The initial condition $a_0 = k_0$ and the transversality condition select the path; goods-market clearing then follows by Walras' law.

Prices as Functions of Capital

To reduce the system, set $A_t = 1$ and substitute $n_t = 1$.

General Technology

$$w_t = f_n(k_t, 1),$$
$$r_t = f_k(k_t, 1) - \delta.$$

Cobb–Douglas Technology

$$w_t = (1-\alpha)k_t^{\alpha},$$
$$r_t = \alpha k_t^{\alpha-1} - \delta.$$

Firm optimality and market clearing remove prices as independent unknowns.

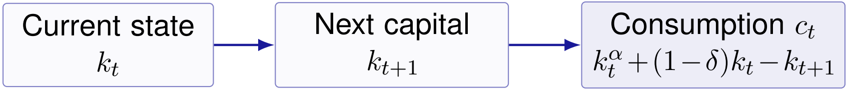
Consumption from Capital

With $n_t = 1$ and $A_t = 1$, the resource constraint is

$$c_t + k_{t+1} = k_t^{\alpha} + (1 - \delta)k_t.$$

Hence,

$$\boxed{c_t = k_t^{\alpha} + (1 - \delta)k_t - k_{t+1}.}$$



Once the capital path is known, the allocation and prices can be recovered.

Three Substitutions Produce the Capital Recurrence

(1) Firm optimality replaces the return:

$$r_{t+1} = \alpha k_{t+1}^{\alpha-1} - \delta.$$

(2) Feasibility replaces consumption at both dates:

$$c_j = k_j^{\alpha} + (1-\delta)k_j - k_{j+1}, \qquad j = t, t+1.$$

(3) Insert both expressions into the household Euler equation:

$$\boxed{\begin{aligned} u'(k_t^{\alpha} + (1-\delta)k_t - k_{t+1}) = \beta \left[1 + \alpha k_{t+1}^{\alpha-1} - \delta\right] \\ \times u'\big(k_{t+1}^{\alpha} + (1-\delta)k_{t+1} - k_{t+2}\big)\,. \end{aligned}}$$

The result is a second-order difference equation in (k_t, k_{t+1}, k_{t+2}).

Necessity and Sufficiency

Necessary Direction

$$\text{interior equilibrium} \quad \Longrightarrow \quad \text{capital recurrence.}$$

Converse Direction

$$\begin{array}{l} \text{capital recurrence} + \text{feasibility} + \text{initial condition} \\ + \text{transversality} + \text{sufficient optimality conditions} \end{array} \quad \Longrightarrow \quad \text{equilibrium.}$$

Neither implication proves that a solution exists or that it is unique.

Pareto Efficiency

Efficiency

A feasible allocation $z^* = (c_t^*, k_{t+1}^*, n_t^*)_{t=0}^{\infty}$ is efficient if there is no feasible $\widetilde{z}$ such that $U(\widetilde{c}) > U(c^*)$.

Efficiency uses only preferences and physical feasibility: no prices and no ownership rule. With one household, the planner selects the efficient path by maximizing the household's lifetime utility over feasible allocations.

The Planner Problem

With $A_t = 1$, the planner solves

$$\max_{(c_t, k_{t+1}, n_t)_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to

$$c_t + k_{t+1} = k_t^{\alpha} n_t^{1-\alpha} + (1-\delta) k_t, \qquad t = 0, 1, \ldots,$$

$$c_t \geq 0, \qquad k_{t+1} \geq 0, \qquad 0 \leq n_t \leq 1, \qquad k_0 > 0 \text{ given.}$$

Since leisure does not enter utility and production increases in labor, $n_t = 1$. At an interior optimum,

$$\lim_{t \to \infty} \beta^t u'(c_t) k_{t+1} = 0.$$

The Planner Euler Equation

For the reduced problem, define

$$L = \sum_{t=0}^{\infty} \beta^t u(c_t) + \sum_{t=0}^{\infty} \lambda_t \left[k_t^{\alpha} + (1-\delta)k_t - c_t - k_{t+1}\right].$$

The first-order conditions give

$$\lambda_t = \beta^t u'(c_t), \qquad t = 0, 1, \ldots,$$

$$\lambda_t = \lambda_{t+1} \left[\alpha k_{t+1}^{\alpha-1} + 1 - \delta\right], \qquad t = 0, 1, \ldots.$$

Therefore,

$$\boxed{u'(c_t) = \beta \left[\alpha k_{t+1}^{\alpha-1} + 1 - \delta\right] u'(c_{t+1}).}$$

Comparing the Two Euler Equations

Competitive Equilibrium

Household Euler equation:

$$u'(c_t) = \beta(1 + r_{t+1})u'(c_{t+1}).$$

Firm return and clearing:

$$1 + r_{t+1} = 1 + \alpha k_{t+1}^{\alpha-1} - \delta.$$

$=$

Social Planner

Planner return:

$$1 + \alpha k_{t+1}^{\alpha-1} - \delta.$$

Planner Euler equation:

$$u'(c_t) = \beta\left[1 + \alpha k_{t+1}^{\alpha-1} - \delta\right]u'(c_{t+1}).$$

The equality is economically powerful only because concavity and transversality make these equations sufficient.

The Market and Planner Choose the Same Allocation

Assume lifetime utility is well-defined, the relevant optima are interior, the problems are concave, and both transversality conditions hold.

Equilibrium $\Longrightarrow$ Planner

Every equilibrium allocation is a planner optimum:

$$\text{equilibrium} \Longrightarrow \text{planner optimum.}$$

Markets attain an efficient allocation.

Planner $\Longrightarrow$ Equilibrium

Every interior planner optimum can be supported by prices:

$$\text{planner optimum} \Longrightarrow \text{equilibrium.}$$

Marginal products construct the prices.

For an infinite horizon, these assumptions are part of the theorem; the Euler equations alone are not sufficient.

Supporting Prices

Start from an interior planner solution $(c_t^*, k_{t+1}^*)_{t=0}^{\infty}$ and define

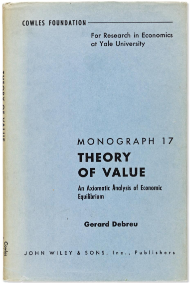
$$w_t^* = (1-\alpha)(k_t^*)^{\alpha}, \qquad r_t^* = \alpha(k_t^*)^{\alpha-1} - \delta.$$

Planner allocation (c^*, k^*) → Marginal-product prices (w^*, r^*) → Household and firm optimize → Markets clear equilibrium

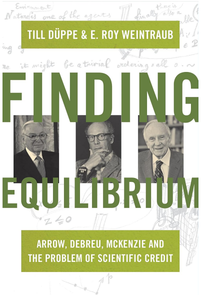
(1) Concavity makes the firm's marginal-product conditions sufficient.
(2) The planner Euler equation and transversality verify household optimality.
(3) Set $a_t^* = k_t^*$ and $n_t^* = 1$; planner feasibility is goods clearing.

Existence of Equilibrium

The definition says what an equilibrium *is*, not that one exists. Existence is a separate theorem: Arrow and Debreu (1954) and, independently, McKenzie (1954), by fixed-point arguments.



Debreu, *Theory of Value* (1959). Cowles Monograph 17.



Düppe and Weintraub, *Finding Equilibrium* (2014).